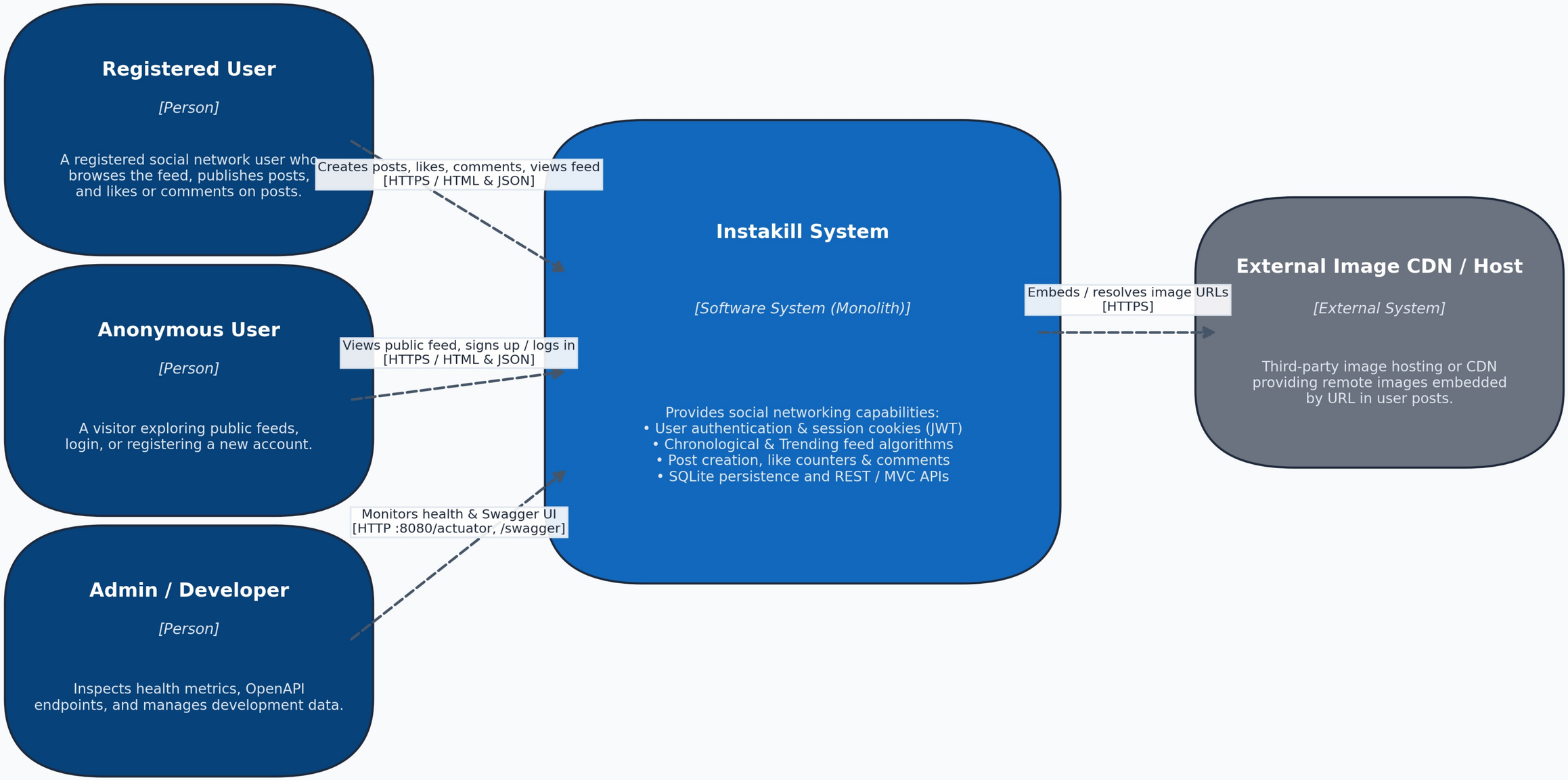
Instakill — System Context Diagram (C4 Level 1)

High-level view of users, external actors, and the Instakill system



Instakill — Container Diagram (C4 Level 2)

Decomposition of the Instakill system into deployable runtime units and technologies

Instakill System Boundary

User

[Person]

Interacts via browser on desktop or mobile devices.

Interacts with SPA
[HTTPS / Browser]

Visits SSR pages
[HTTP :8080 / HTML]

Loads images directly in browser
[HTTPS]

Frontend SPA

[Container: Vue 3, Vite, Axios]

Single-page application delivering an Instagram-like reactive feed, composer, likes toggle, and JWT cookie auth.
(Port :5173)

REST API calls with JWT cookie
[JSON / HTTP Proxy :8080]

Server-Side MVC UI

[Container: Thymeleaf, Spring MVC]

Server-rendered HTML templates for feed viewing and authentication fallback.
(Port :8080/feed, /auth/login)

Internal MVC routing
[In-process Spring Dispatcher]

Backend Application

[Container: Spring Boot 3.3.2, Java 21]

Exposes REST endpoints (/api/**) & MVC views.
Contains business domains: User, Post, Feed (Strategy), Interaction (Event Publisher).
Secured by Spring Security 6 & JWT cookies.

Queries & Transactions
[JDBC / SQL / HikariCP]

Database

[Container: SQLite 3]

Single-file relational database (instakill.db) storing accounts, posts, likes, comments.
Accessed through SQLite-JDBC & Hibernate.

External Image CDN

[External System]

Hosts remote media linked inside post imageUrl attributes.

Instakill — Backend Component Diagram (C4 Level 3)

Detailed internal structure of the Spring Boot 3 Backend Container

